# Supervised Principal Component Analysis and LASSO Multinomial Logistic Regression for Multiclass Cancer Classification: A Comparative Study on RNA-Seq Gene Expression Data


## Abstract

High-dimensional gene expression data from RNA sequencing (RNA-Seq) platforms present significant analytical challenges because the number of features is much larger than the number of observations, a condition known as the $p \gg n$ problem. This study compares two dimensionality reduction and classification approaches: Supervised Principal Component Analysis (SPCA) combined with multinomial logistic regression and LASSO multinomial logistic regression. The analysis used the TCGA PANCAN RNA-Seq dataset, consisting of 801 samples, 20,531 genes, and five cancer types: BRCA, COAD, KIRC, LUAD, and PRAD. All features were standardised prior to analysis, and a stratified 70:30 train-test split was applied for unbiased evaluation. SPCA screened genes using one-way ANOVA F-statistics, selected the top 100 genes via 5-fold cross-validation, and extracted six principal components explaining 90.42% of the variance. LASSO selected 33 genes with non-zero coefficients through 10-fold cross-validation at $\lambda = 0.02344$. Both methods achieved identical and perfect classification performance on the test set, with 100% overall accuracy, a macro F1-score of 1.0000, and a multiclass AUC of 1.0000 across all cancer types. While predictive performance was equivalent, the methods differed in feature reduction strategy: SPCA produced a more compact six-dimensional model input, whereas LASSO directly identified 33 interpretable gene-level predictors. These findings suggest that both methods are highly effective for RNA-Seq cancer classification, with the preferred choice depending on whether compact representation or gene-level interpretability is prioritised.



**Keywords:** supervised principal component analysis, LASSO, multinomial logistic regression, RNA-Seq, cancer classification, dimensionality reduction


## 1. Introduction

Cancer is one of the leading causes of death around the world and represents a major global health challenge. According to the World Health Organization, cancer was responsible for nearly 10 million deaths globally in 2022, making it the second leading cause of mortality after cardiovascular diseases (World Health Organization, 2022). Early detection and accurate classification of cancer types are essential for improving treatment strategies and patient outcomes. Cancer heterogeneity contributes to differences in treatment response, disease progression, and prognosis among patients. This heterogeneity reflects distinct genetic, molecular, and cellular characteristics across cancer types and subtypes (Alanazi et al.,2024). In gene expression studies, these molecular differences can be observed through RNA expression patterns and used to support cancer classification. For example, multiclass cancer datasets may include several tumor types, such as BReast invasive CArcinoma (BRCA), KIdney Renal clear cell Carcinoma (KIRC), COlon ADenocarcinoma (COAD), LUng

ADenocarcinoma (LUAD), and PRostate ADenocarcinoma (PRAD). This multiclass structure requires classification methods that can distinguish among more than two cancer categories.

Based on these molecular differences, gene expression profiling has become an important approach for studying cancer. Among different omics levels, transcriptomics has been widely used, and the development of next-generation sequencing has made RNA sequencing (RNA-Seq) a mainstream platform for gene expression analysis (Alanazi et al., 2024). Compared with earlier microarray-based techniques, RNA-Seq represents a major advancement in gene expression analysis because it enables unbiased transcript detection, provides higher sensitivity, and allows the quantification of both known and novel transcripts (Antoszewski et al., 2022). The ability to measure the expression levels of thousands of genes simultaneously offers a comprehensive molecular picture of tumor biology. In cancer research, RNA-Seq has been increasingly used to identify biomarkers, discover molecular subtypes, and develop predictive models for clinical outcomes (Wang et al., 2009).

Despite its advantages, gene expression data from RNA-Seq platforms present significant analytical challenges. Gene expression datasets are typically characterized by high dimensionality, where the number of variables parameters $p$ greatly exceeds the number of observations $n$, a condition known as the "$p \gg n$" problem. This condition may lead to overfitting, computational instability, and reduced predictive performance in classification models (Fan & Lv, 2010). The curse of dimensionality becomes more severe as the number of irrelevant or redundant features increases. In a typical RNA-Seq dataset, there may be more than 20,000 gene expression measurements per sample, but only a few hundred samples are available for analysis. When all genes are used directly in a classification model, the model may fit noise instead of the true biological signal, resulting in poor generalization to new data (Hastie et al., 2009). Appropriate dimensionality reduction and feature selection methods are needed before building classification models.

Due to the large number of correlated variables in high-dimensional gene expression datasets, Principal Component Analysis (PCA) is commonly used as a dimensionality reduction technique. PCA transforms the original variables into a smaller number of new variables called principal components (PCs). These components are uncorrelated with each other and are ordered according to the amount of variance they explain. In this way, the first few components can represent most of the important information in the original data. Therefore, PCA makes high-dimensional data easier to analyze while still preserving the main variability in the dataset (Jolliffe, 2002). Given a data matrix $X$ with dimensions $n \times p$, where $n$ represents the number of observations and $p$ represents the number of genes, PCA transforms the original variables into a new orthogonal space. Each principal component is formed as a linear combination of the original features. The components are ordered according to the amount of variance they explain, with the first principal component capturing the largest variance in the data (Hastie et al., 2009). By selecting only the first few principal components, the dimensionality of the data can be greatly reduced while retaining most of the information from the original dataset.

However, conventional PCA is an unsupervised method. This means that it does not use information about the response variable, such as cancer class labels, when constructing the principal components. The components with the largest variance are not necessarily the most relevant for predicting cancer classes. In fact, variation in gene expression may be driven by biological processes unrelated to cancer type, such as batch effects, age, or environmental factors. As a result, the principal components that explain the most variance may not be the most useful for classification (Bair & Tibshirani, 2006). To overcome this limitation, Bair et al. (2006) proposed Supervised Principal Component Analysis (SPCA). Unlike conventional PCA, SPCA incorporates information from the response variable during the feature screening stage before constructing principal components. Instead of applying PCA to all variables, SPCA selects features that are strongly associated with the response variable and then performs PCA only on the selected subset. This approach is expected to produce components that are more relevant for prediction and classification tasks.

Unlike conventional PCA-based approaches that perform unsupervised dimensionality reduction prior to classification (e.g., Tunny et al., 2024), SPCA incorporates class label information during the feature screening stage, thereby producing principal components that are more relevant to cancer classification. The SPCA procedure in this study follows the framework of Bair et al. (2006). In their original framework, univariate regression coefficients are used for feature screening. This study adapts the framework by using the F-statistic from one-way ANOVA as the screening criterion for categorical response variables. Genes with the largest F-statistics are selected, PCA is applied to this reduced set, and the extracted principal components are used as predictors in a multinomial logistic regression model.

In addition to SPCA, the Least Absolute Shrinkage and Selection Operator (LASSO) is used as a comparison method. LASSO was introduced by Tibshirani (1996) as a regularization technique for regression models. The LASSO method performs simultaneous coefficient shrinkage and variable selection by adding an $L_1$ penalty on the regression coefficients. This penalty forces some of the coefficients to be exactly zero, effectively removing the corresponding variables from the model. The amount of shrinkage is controlled by a tuning parameter, typically selected using cross-validation (Tibshirani, 1996; Hastie et al., 2015). This property makes LASSO particularly useful for high-dimensional gene expression datasets, where the number of genes far exceeds the number of samples. LASSO multinomial logistic regression extends the LASSO framework to multiclass problems by applying the $L_1$ penalty to the coefficients of each logit function (Friedman et al., 2010).

Several studies have compared dimension reduction and regularization methods for high-dimensional classification. However, limited research has focused specifically on comparing SPCA with LASSO in the context of multiclass cancer classification using RNA-Seq data. SPCA combines the interpretability of principal components with supervised feature screening, while LASSO directly performs feature selection within the classification model. Each method has its own strengths and limitations. SPCA reduces dimensionality before modeling, which can reduce computational burden and stabilize estimation. LASSO integrates feature selection and classification in a single step, which may lead to more parsimonious models. A direct comparison of these two approaches using the same dataset and evaluation criteria can provide valuable insights for researchers working on high-dimensional cancer classification.

This study aims to compare supervised principal component analysis and LASSO multinomial logistic regression for high-dimensional cancer classification using RNA-Seq gene expression data. The comparison focuses on classification performance and the effectiveness of feature reduction in multiclass cancer prediction. By evaluating both methods on a real high-dimensional multiclass dataset, this study aims to provide practical guidance for selecting appropriate classification strategies in genomic studies.

## 2. Problem Definition

This study addresses the problem of high-dimensional multiclass cancer classification using RNA-Seq gene expression data. The dataset contains a very large number of gene features, while the number of samples is relatively small. This condition can cause overfitting, computational instability, and difficulty in identifying informative genes for classification.

The main problem is how to reduce the dimensionality of gene expression data while maintaining good classification performance. In this study, two methods are compared: Supervised Principal Component Analysis (SPCA) and LASSO multinomial logistic regression. SPCA reduces dimensionality by selecting genes related to the cancer class and then constructing principal components, while LASSO performs feature selection by shrinking some gene coefficients to zero.

Therefore, this study aims to determine which method provides better performance for multiclass cancer classification and which method is more effective in reducing the number of gene features.

## 3. Research Questions and Hypotheses

### 3.1 Research Questions

This study is guided by four research questions addressing the core comparison between Supervised Principal Component Analysis (SPCA) and LASSO multinomial logistic regression for high-dimensional cancer classification:

1) How well does SPCA perform in multiclass cancer classification using RNA-Seq gene expression data?
2) How well does LASSO multinomial logistic regression perform in multiclass cancer classification using RNA-Seq gene expression data?
3) Which method provides better classification performance between SPCA and LASSO?
4) Which method is more effective in reducing the number of gene features?

### 3.2 Hypotheses

This study evaluates the two methods based on two comparative propositions rather than formal statistical hypotheses. The main goal is to compare the methods descriptively, not to conduct significance testing.

1) **Classification Performance**
   Both SPCA and LASSO are expected to achieve high classification accuracy on the test set because RNA-Seq gene expression data contain strong molecular signals for separating cancer types. The comparison focuses on whether the two methods differ in overall accuracy, macro F1-score, and multiclass AUC. If both methods produce the same performance, neither method is considered better for prediction.
2) **Feature Reduction**
   LASSO is expected to select fewer original genes than SPCA because it performs direct gene selection through coefficient shrinkage. This may be useful in practical diagnostic settings, as a smaller number of genes can reduce the number of variables that need to be analysed and may support a simpler and more interpretable classification workflow.

## 4. Dataset Description

The dataset used in this study was obtained from the UCI Machine Learning Repository. It consists of $n = 801$ observations and $p = 20{,}531$ gene expression features, measured via RNA-Seq, creating a high-dimensional setting where $p \gg n$. The response variable consists of five cancer classes ($K = 5$): BRCA, KIRC, COAD, LUAD, PRAD. There are no missing values in the dataset.

## 5. Methodology

This section describes the complete analytical pipeline used to compare Supervised Principal Component Analysis (SPCA) and LASSO multinomial logistic regression for multiclass cancer classification. The pipeline consists of five steps: (1) data preprocessing, (2) train-test split, (3) SPCA procedure, (4) LASSO multinomial logistic regression, and (5) model evaluation. The R packages used are caret, nnet, glmnet, and pROC.

### 5.1 Data Preprocessing and Standardisation

Before the analysis, all $p = 20{,}531$ gene expression features are standardised to zero mean and unit variance. This step is essential because PCA is sensitive to the scale of variables, and the $L_1$ penalty in LASSO penalises all coefficients equally only when predictors are on the same scale. For each gene $j = 1, \ldots, p$ and observation $i = 1, \ldots, n$, the standardised value is:

$$x^*_{ij} = \frac{(x_{ij} - \mu_j)}{\sigma_j} \tag{1}$$

where $\mu_j$ and $\sigma_j$ are the mean and standard deviation of gene j, computed from the training set only. The same parameters $\mu_j$ and $\sigma_j$ are then applied to standardise the test set, preventing data leakage. In R, this is implemented using preProcess(method = c("center", "scale")) from the caret package.

### 5.2 Train-Test Split

The dataset of $n = 801$ observations is partitioned into a training set and a test set using stratified random sampling in a 70:30 ratio, producing $n_{train} = 563$ training observations and $n_{test} = 238$ test observations. Stratified sampling was used to keep the proportion of the five cancer classes similar in both datasets. All model training, feature selection, and hyperparameter tuning are performed exclusively on the training set. The test set is held out and used only once for the final performance evaluation. In R, stratified splitting is performed using createDataPartition() from the caret package with set.seed(42) for reproducibility.

### 5.3 Supervised Principal Component Analysis (SPCA)

The SPCA procedure is implemented manually following the framework of Bair et al. (2006). Unlike using a dedicated package, each step is coded explicitly in R to ensure full control over the feature screening threshold and the number of principal components. The procedure consists of four steps.

**Step 1: Univariate Gene Screening via ANOVA F-Statistic**

For each gene $j = 1, \ldots, p$, a one-way analysis of variance (ANOVA) F-statistic is calculated to measure how strongly the gene is related to the cancer classes *(BRCA, KIRC, COAD, LUAD, PRAD)*. The F-statistic is defined as the ratio of the between-group variance (MSB) to the within-group variance (MSW):

$$F_j = \frac{MSB_j}{MSW_j} \tag{2}$$

where:

$$MSB_j = \frac{1}{K-1} \sum_{k=1}^{K} n_k \, (\bar{x}_{kj} - \bar{x}_j)^2 \tag{3}$$

$$MSW_j = \frac{1}{n_{\text{train}} - K} \sum_{k=1}^{K} \sum_{i \in C_k} (x_{ij} - \bar{x}_{kj})^2 \tag{4}$$

with $K = 5$ is the number of cancer classes, $n_k$ the number of observations in class $k$, $\bar{x}_{kj}$ the mean expression of gene $j$ in class $k$, and $\bar{x}_j$ the overall mean of gene $j$. A large $F_j$ indicates that gene $j$ has high discriminative power across cancer classes. The F-statistics were computed manually using the between-group and within-group mean squares for each gene.

**Step 2: Gene Ranking and Selection of Top-k Genes**

After calculating the F-statistic for each gene, all $p = 20{,}531$ genes are ranked from the highest to the lowest F-statistic value. Genes with higher F-statistic values are considered more useful for separating cancer classes. The top $k$ genes are then selected as the candidate feature set:

$$S(k) \;=\; \{\, j : F_j \; is\ among\ the\ k\ largest\ F - statistics \,\} \tag{5}$$

In this study, three values of $k$are considered: $k \in \{100, 500, 1000\}$. The best value of $k$is selected using 5-fold cross-validation on the training set. The value that gives the highest mean cross-validation accuracy is chosen. This approach is easier to interpret because it directly compares different numbers of selected genes.

**Step 3: PCA on Selected Genes**

After the top-$k$ genes are selected, PCA is applied only to the reduced training matrix:

$$X_{S(k)} \in \mathbb{R}^{n_{train} \times k} \tag{6}$$

where $X_{S(k)}$contains the selected $k$ genes. PCA transforms these selected genes into a smaller number of principal components. PCA is calculated using Singular Value Decomposition (SVD):

$$X\{S(k)\} \;=\; \boldsymbol{U}\,\boldsymbol{\Sigma}\,\boldsymbol{V}^{\mathrm{T}} \tag{7}$$

where $U$ contains the left singular vectors, $\mathbf{\Sigma}$ is a diagonal matrix of singular values $\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_k \geq 0$, and $V$ contains the right singular vectors (loading vectors). The $m$-th principal component score is:

$$z_{\mathrm{m}} \;=\; X_{S(k)}\, v_{\mathrm{m}}\,, \quad m \;=\; 1, 2, \ldots, K_{Pc} \tag{8}$$

where $v_m$ is the $m$-th column of $V$. The number of principal components $K_{Pc}$ retained is determined by the cumulative proportion of variance explained (PVE):

$$PVE_{(m)} = \frac{\sum_{\ell=1}^{m} \sigma_\ell^2}{\sum_{\ell=1}^{k} \sigma_\ell^2} \tag{9}$$

Components are retained until $PVE_{(m)} \geq 0.90$, meaning that the selected components explain at least 90% of the total variance in the selected gene expression matrix. At least two principal components are retained. For the test set, the PC scores are computed by projecting the test data onto the same loading vectors $V_{K_{PC}}$ estimated from the training set:

$$Z_{\text{test}} = X_{\text{test},S(k)} V_{K_{PC}} \tag{10}$$

In R, PCA is performed using prcomp(center = FALSE, scale. = FALSE) since the data are already standardised. Test set scores are obtained via predict(pca_model, newdata = X_test).

**Step 4: Multinomial Logistic Regression on PC Scores**

The retained PC scores $Z_{\text{train}} \in \mathbb{R}^{n_{\text{train}} \times K_{PC}}$ are used as predictors in a multinomial logistic regression model. For observation $i$ with PC score vector $z_i = (z_{i1}, z_{i2}, \dots, z_{iK_{PC}})^T$, the model estimates the probability of belonging to cancer class *k* as:

$$P(y_i = k \mid z_i) = \frac{\exp(\beta_{0k} + z_i^T \beta_k)}{\sum_{\ell=1}^{K} \exp(\beta_{0\ell} + z_i^T \beta_\ell)} \tag{11}$$

where $\beta_{0k}$ is the intercept and $\beta_k \in$ ℝ^{K_PC} is the coefficient vector for class *k*. One class is set as the reference category with all coefficients fixed to zero for identifiability. The model parameters are estimated by maximising the multinomial log-likelihood:

$$\ell(\beta) = \sum_{i=1}^{n_{\text{train}}} \sum_{k=1}^{K} \mathbf{1}(y_i = k) \log P(y_i = k \mid z_i) \tag{12}$$

The predicted cancer class for a new observation is:

$$\hat{y}_i = \arg \max_{k \in \{1, \dots, K\}} P(y_i = k \mid z_i) \tag{13}$$

In R, multinomial logistic regression is fitted using multinom(y ~ ., data = df_train, trace = FALSE) from the nnet package.

## 5.4 LASSO Multinomial Logistic Regression

As a comparison method, LASSO multinomial logistic regression is applied directly to the full standardised gene expression matrix $X_{\text{train}} \in \mathbb{R}^{n_{\text{train}} \times p}$ without prior dimensionality reduction. LASSO (Tibshirani, 1996) imposes an $L_1$ penalty on the regression coefficients, simultaneously performing variable selection and coefficient estimation. The objective function minimised is:

$$\hat{\beta} = \arg \min_{\beta} \left\{ -\frac{1}{n_{\text{train}}} \ell(\beta) + \lambda \sum_{j=1}^{p} \sum_{k=1}^{K-1} |\beta_{jk}| \right\} \tag{14}$$

where $\ell(\beta)$ is the multinomial log-likelihood, $\beta_{jk}$ is the coefficient of gene *j* for class *k*, and $\lambda \geq 0$ is the regularisation parameter. The $L_1$ penalty has the property of shrinking many coefficients exactly to zero, so that only a subset of genes receives non-zero coefficients effectively performing automatic gene selection. The set of genes selected by LASSO is:

$$S(\lambda) = \{ j : \beta_{jk} \neq 0 \text{ for at least one class } k \} \tag{15}$$

The regularisation parameter $\lambda$ is selected using 10-fold cross-validation on the training set, minimising the cross-validated misclassification error. The optimal value

*λ_min* — the value that minimises the CV error — is used for the final model. In R, this is implemented using:
cv.glmnet(x = X_train, y = y_train, family = "multinomial", alpha = 1, nfolds = 10, type.measure = "class")
The predicted class probabilities for a new observation $x_i$ are:

$$P(y_i = k \mid x_i) = \frac{exp\,(\beta_{0k}+x_i^T\beta_k)}{\sum_{\ell=1}^{K} exp\,(\beta_{0\ell}+x_i^T\beta_\ell)} \tag{16}$$

and the predicted class is $\hat{y}_i = arg\ max\_k\ P(y_i = k \mid x_i)$. Predictions on the test set are obtained using predict(lasso_cv, newx = X_test, s = "lambda.min", type = "class").

### 5.5 Model Evaluation

Both models are evaluated on the held-out test set ($n_{\text{test}} = 238$). The evaluation metrics include overall accuracy, sensitivity, specificity, precision, macro-averaged F1-score, and multiclass AUC. These metrics are used to compare the classification performance of SPCA and LASSO multinomial logistic regression. For each cancer class $k$, let $TP_k$, $TN_k$, $FP_k$, and $FN_k$ denote the true positives, true negatives, false positives, and false negatives respectively.

1) Overall Accuracy

$$Accuracy = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \mathbf{1}(\hat{y}_i = y_i) \tag{17}$$

2) Sensitivity (Recall) per class k

$$Sensitivity_k = \frac{TP_k}{TP_k+FN_k} \tag{18}$$

3) Specificity per class k

$$Specificity_k = \frac{TN_k}{TN_k+FP_k} \tag{19}$$

4) Precision per class k

$$Precision_k = \frac{TP_k}{TP_k+FP_k} \tag{20}$$

5) Macro-averaged F1-Score

$$F1 = \frac{1}{K}\sum_{k=1}^{K} \frac{2\times Precision_k \times Sensitivity_k}{Precision_k + Sensitivity_k} \tag{21}$$

6) Multiclass AUC

For multiclass problems with $K > 2$ classes, the standard binary AUC is extended using the Hand-Till (2001) method, which averages the pairwise AUC values across all $\frac{K(K-1)}{2} = 10$ class pairs:

$$AUC = \frac{2}{K(K-1)} \sum_{k<l} A(k,l) \tag{22}$$

where $A(k,l)$ is the pairwise AUC between class $k$ and class $l$. Since this study has $K = 5$ cancer classes, there are 10 pairwise class comparisons. Multiclass AUC is computed using the `multiclass.roc()` function from the `pROC` package. For feature reduction, SPCA is reported using the number of selected genes $k$and the number of retained principal components $K_{PC}$. LASSO is reported using the number of genes with at least one non-zero coefficient, denoted by $\mid S(\lambda_{\min}) \mid$.

## 6. Model Fitting and Evaluation

This section describes how the SPCA and LASSO models were fitted and evaluated. The process includes feature selection, dimensionality reduction, and hyperparameter tuning. Both models were trained using only the training set (n = 563), while performance was assessed on an independent test set (n = 238) to provide an unbiased evaluation of classification performance.

### 6.1 SPCA: Gene Screening and Principal Component Selection

The SPCA pipeline began with univariate gene screening using one-way ANOVA F-statistics computed across the five cancer classes. These statistics were used to rank the genes according to how strongly their expression levels differed across the five cancer classes. To determine the optimal number of selected genes, a 5-fold cross-validation was performed on the training set using $k \in \{100,500,1000\}$. For each fold, the ANOVA F-statistics were recalculated using only the fold-specific training data. The top-$k$ genes were then selected, PCA was applied, and a multinomial logistic regression model was fitted using the resulting PC scores. The mean validation accuracy across the five folds was recorded for each value of $k$.

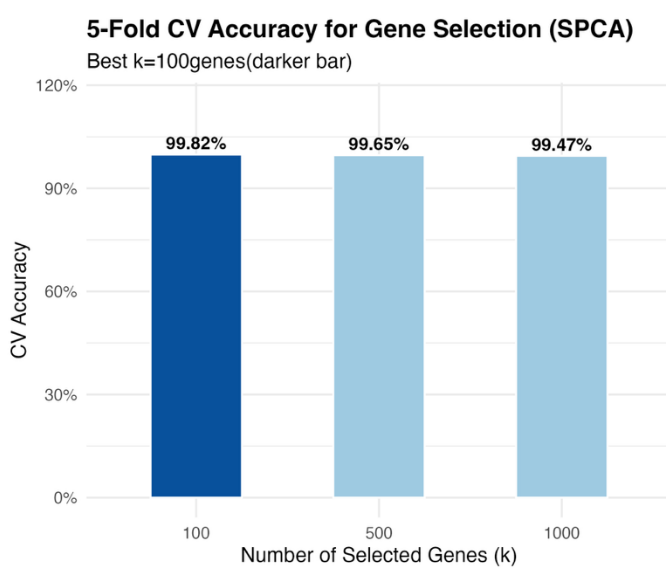


**Figure 1.** 5-Fold CV Accuracy for Gene Selection (SPCA)

As shown in Figure 1, k = 100 achieved the highest 5-fold CV accuracy of 99.82%, followed by k = 500 (99.65%) and k = 1000 (99.47%). Although all three values yielded high accuracy, k = 100 was selected as the optimal threshold because it achieved the best cross-validated performance while also using the fewest genes, favouring parsimony.

PCA was then applied to the top-100 gene expression matrix ($n_{train} \times 100$). The cumulative proportion of variance explained (PVE) was used to determine the number of principal components to retain. Figure 2 presents the scree plot from this PCA.

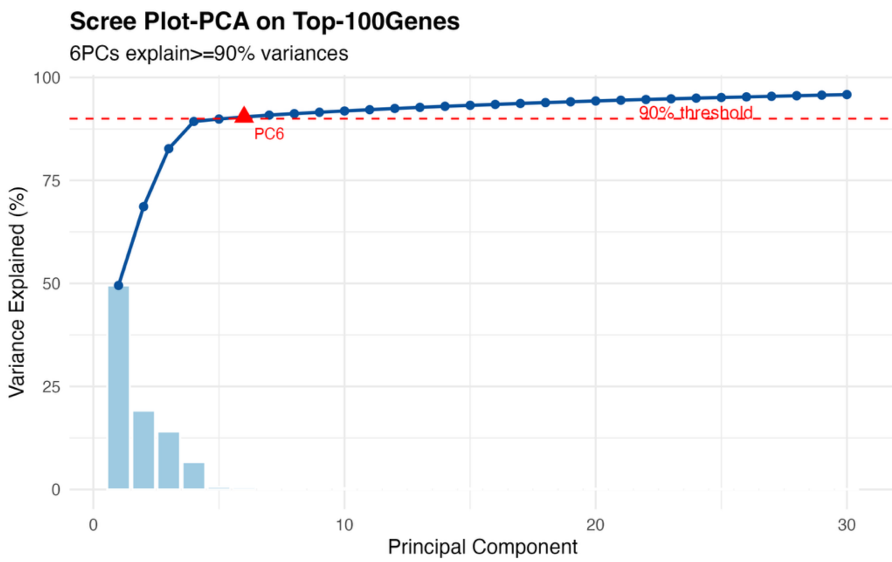


**Figure 2**. Scree Plot — PCA on Top-100 Genes

The scree plot shows that the first principal component alone accounts for approximately 49.5% of the variance, and the cumulative PVE crosses the 90% threshold at PC6 (cumulative PVE = 90.42%). Therefore, m = 6 principal components were retained. These 6 PC scores were used as predictors in the multinomial logistic regression model, reducing the feature space from 20,531 genes to 6 principal component scores.

A multinomial logistic regression model was then fitted on the 6 PC scores using the nnet::multinom() function in R. The reference class was set automatically, and model parameters were estimated by maximising the multinomial log-likelihood. No additional regularisation was applied, as the PC scores are uncorrelated and far fewer in number than observations.

### 6.2 LASSO: Lambda Selection and Gene Identification

LASSO multinomial logistic regression was applied directly to the full standardised training matrix $X_{\text{train}} \in \mathbb{R}^{563\times 20531}$. The regularisation parameter $\lambda$was selected using 10-

fold cross-validation by minimising the cross-validated misclassification error. Figure 3 shows the cross-validation curve for different values of $\lambda$.

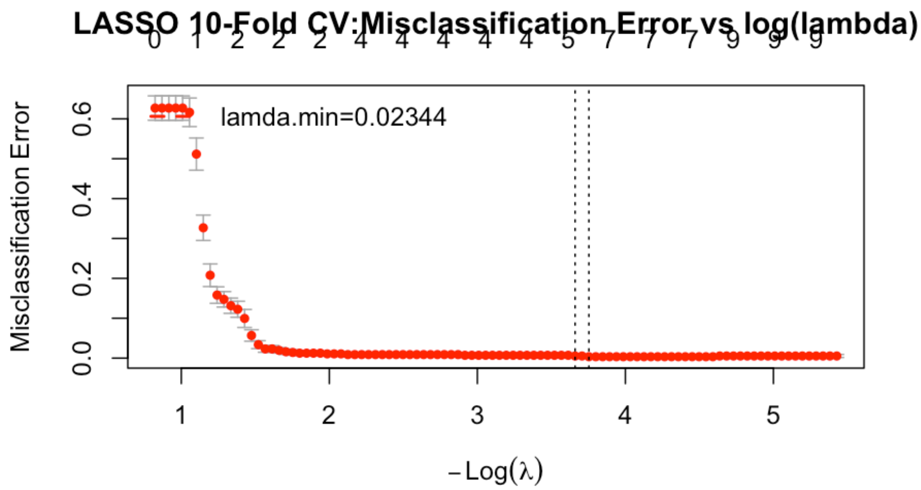


**Figure 3.** LASSO 10-Fold CV : Misclassification Error vs. log(λ)

The CV curve shows that the misclassification error decreases as the strength of regularisation is reduced and reaches its minimum at $\lambda_{min} = 0.02344$. At this optimal value, LASSO retained 33 genes with at least one non-zero coefficient across the five cancer classes. These 33 genes were used as the final gene-level predictors, representing a 99.84% reduction from the original 20,531 genes.

## 7. Results

Both SPCA and LASSO multinomial logistic regression were evaluated on the held-out test set (n = 238) using overall accuracy, macro-averaged F1-score, multiclass AUC, and per-class sensitivity, specificity, and precision. This section presents the results of both methods in terms of classification performance and feature reduction.

### 7.1 Classification Performance

**Table 1.** Overall Classification Metrics on the Test Set

| Metric | SPCA + Multinomial LogReg | LASSO Multinomial LogReg |
|---|---|---|
| Overall Accuracy | 1.0000 (100.00%) | 1.0000 (100.00%) |
| Macro F1-Score | 1.0000 | 1.0000 |
| Multiclass AUC | 1.0000 | 1.0000 |

Table 1 summarises the overall classification metrics for both models on the test set. Both models achieved perfect classification performance on all three global metrics: 100% accuracy, a macro F1-score of 1.0000, and a multiclass AUC of 1.0000. This result indicates that neither model misclassified a single test observation across all five cancer types. The confusion matrices for both models are presented in Figures 4 and 5. Both matrices show all predictions falling on the main diagonal, with zero off-diagonal entries, confirming the absence of any misclassification.

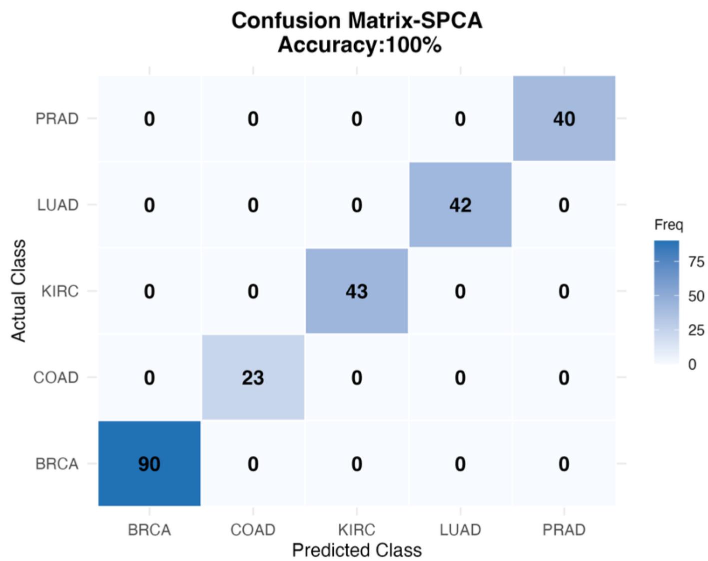


**Figure 4.** Confusion Matrix SPCA + Multinomial Logistic Regression (Test Set)

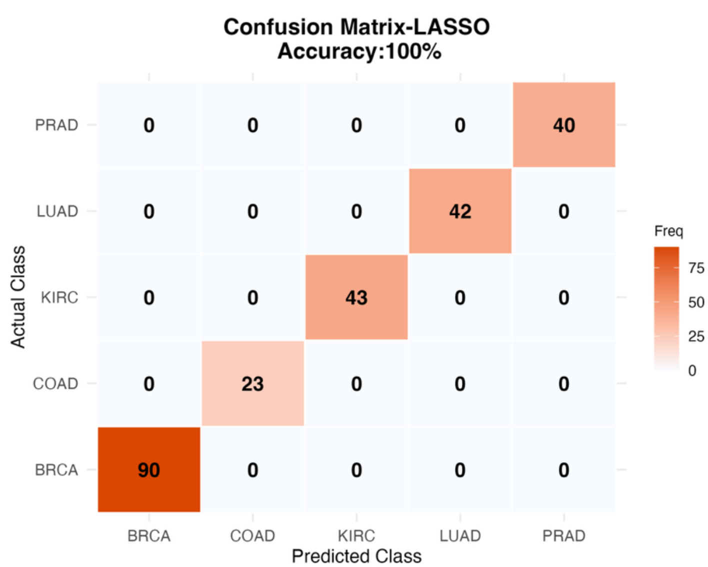


**Figure 5.** Confusion Matrix — LASSO Multinomial Logistic Regression (Test Set)

Table 2 presents the per-class performance metrics for both models. Sensitivity, specificity, precision, and F1-score are all equal to 1.0000 for every cancer class under both methods.

**Table 2.** Per-Class Performance Metrics — SPCA and LASSO (Test Set)

| Cancer Class | Sensitivity | Specificity | Precision | F1-Score | n_test |
|---|---|---|---|---|---|
| BRCA | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 90 |
| COAD | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 23 |
| KIRC | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 43 |
| LUAD | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 42 |
| PRAD | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 40 |

Per-class metrics are identical for both SPCA and LASSO, as both models produced zero misclassifications on the test set. The result holds consistently across all five cancer types, including COAD (n = 23), which has the smallest test-set representation. Figure 6 presents the PCA score plot of the training data projected onto PC1 and PC2, coloured by cancer type. This visualisation illustrates the degree of class separability achieved in the supervised PC space.

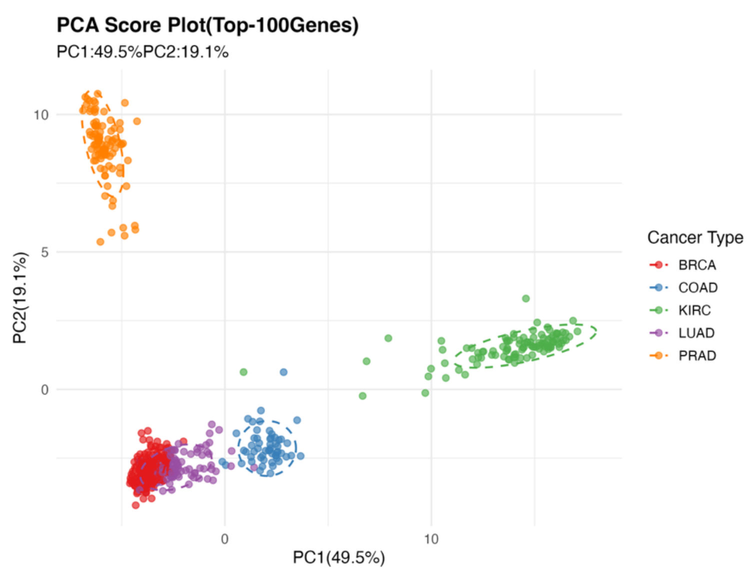


**Figure 6.** PCA Score Plot — PC1 vs. PC2 (Top-100 Genes)

The score plot shows that the SPCA procedure creates a PC space with clear class separation. KIRC is located far to the right along PC1, while PRAD forms a separate cluster in the upper-left area. BRCA, LUAD, and COAD are located in the lower-left area, with COAD slightly separated from the other two classes. Although the first two PCs do not show the full information, the separation seen in PC1 and PC2, together with the remaining four PCs, helps the multinomial logistic regression model classify all test observations correctly.

Figure 7 illustrates the F1-score per cancer class for both methods side by side, confirming equivalent per-class performance. Both methods achieve F1 = 1.00 for all five cancer types, with no observable difference in per-class performance.

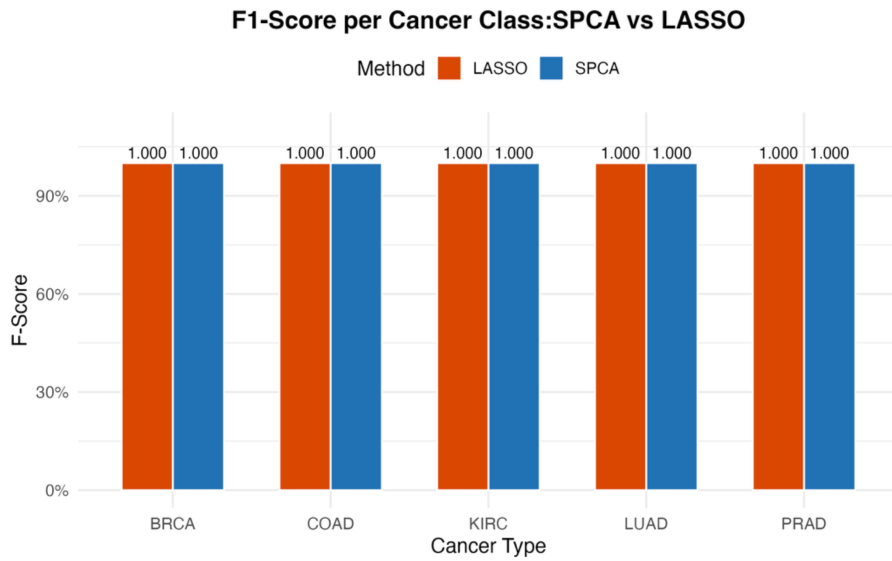


**Figure 7.** F1-Score per Cancer Class: SPCA vs. LASSO

### 7.2 Feature Reduction

While both models achieved identical classification performance, they differ substantially in the number of features used and the degree of dimensionality reduction. Table 3 summarises the feature reduction characteristics of each method.

**Table 3.** Feature Reduction Summary

| Method | Total Genes | Selected Genes | Final Features (Model Input) | Reduction (%) |
|---|---|---|---|---|
| SPCA | 20,531 | 100 (top-k by ANOVA F-stat) | 6 (PC scores) | 99.97% |
| LASSO | 20,531 | 33 (non-zero coefficient genes) | 33 (gene coefficients) | 99.84% |

SPCA reduced the original 20,531 gene expression matrix in two stages. First, 100 genes were selected using ANOVA F-statistic screening. Then, PCA compressed these genes into 6 principal component scores, which were used as the final model inputs. In contrast, LASSO directly selected 33 genes from the full feature space and used these genes as predictors. Therefore, LASSO selected fewer original genes, whereas SPCA produced a more compact final model input. Overall, both methods reduced the original feature space by approximately 99.8–99.9%.

### 7.3 Research Question Evaluation

This study does not perform formal statistical hypothesis testing because the main goal is descriptive comparison. Instead, the research questions are evaluated based on the observed test-set metrics. Regarding classification performance (RQ1–RQ3), both SPCA and LASSO achieved identical results across all metrics: 100% accuracy, a macro F1-score of 1.0000, and a multiclass AUC of 1.0000. Therefore, no difference in predictive performance was observed between the two methods on this dataset.

Regarding feature reduction (RQ4), the two methods showed different reduction patterns. SPCA selected (k = 100) genes using ANOVA screening and then compressed them into 6 principal components as the final model inputs. LASSO selected 33 genes with non-zero coefficients and used these genes directly as predictors. Thus, SPCA produced a more compact final model input, while LASSO provided a more directly interpretable gene-level feature set. Therefore, although the two methods achieved the same predictive performance, they differed in how they reduced the original feature space.

## 8. Discussion

Both SPCA and LASSO multinomial logistic regression achieved perfect classification performance on the test set. Both methods obtained 100% accuracy, a macro F1-score of 1.0000, and a multiclass AUC of 1.0000 for all five cancer types. This result is consistent with Alanazi et al. (2024), who reported high classification accuracy using machine learning methods on RNA expression data. It also shows that the five cancer types in the TCGA PANCAN dataset have strong gene expression differences. The PCA score plot (Figure 6) shows clear separation among several cancer classes, especially KIRC and PRAD. Together with the other retained PCs, this separation helps explain why both methods classified all test samples correctly.

One important difference between SPCA and ordinary PCA is that SPCA uses class-label information during gene screening. Ordinary PCA is unsupervised, so it does not use cancer class labels when forming principal components. Therefore, the first principal components may not always be the most useful for classification. In this study, SPCA first selected genes using ANOVA F-statistics and then applied PCA only to the selected genes. This step helps make the PC scores more relevant for cancer classification. Similar dimensionality reduction methods were used by Tunny et al. (2024), but direct comparisons between SPCA and LASSO multinomial logistic regression on this dataset are still limited.

Although both methods produced the same classification performance, they reduced the feature space in different ways. LASSO selected 33 genes with non-zero coefficients and used these genes directly as predictors. This makes LASSO easier to interpret at the gene level. In contrast, SPCA first selected 100 genes and then compressed them into 6 principal components. This makes the final SPCA model more compact, but the PC scores are harder to interpret biologically because each PC is a combination of many genes. Therefore, LASSO is more suitable when the goal is to identify specific genes, while SPCA is more suitable when the goal is to build a compact prediction model.

Overall, the results show that both SPCA and LASSO are effective for multiclass cancer classification using high-dimensional RNA-Seq data. Neither method performed better in terms of predictive accuracy, but each method has a different advantage. LASSO provides better gene-level interpretability, while SPCA provides a smaller final model input.

## 9. Limitations and Risks

Some limitations of this study are discussed in this section. First, the evaluation used only one stratified 70:30 train-test split with a fixed random seed. This makes the results reproducible and keeps the class proportions balanced, but the performance may still change slightly if a different random split is used. Future studies could use repeated cross-validation or repeated train-test splits to get more stable performance results.

Second, the TCGA PANCAN dataset includes five cancer types from different tissue origins. These cancer types have clear differences in their gene expression profiles. This may be one reason why both SPCA and LASSO achieved perfect classification performance. Because of this, the results may not fully apply to more difficult classification tasks, such as separating molecular subtypes within the same tissue type or working with datasets where the classes are more similar to each other.

Another limitation is related to interpretability. LASSO directly selects a group of genes, so the selected genes can be studied one by one. In contrast, SPCA produces principal components, which are combinations of several genes. This makes SPCA harder to interpret from a biological point of view. Further analysis, such as examining component loadings or studying biological pathways, would be needed to understand how individual genes contribute to the classification results.

Future work could expand this comparison to other cancer datasets, test the methods using repeated resampling approaches, and explore the biological meaning of the selected genes and principal components.

## 10. Conclusion

This study compared Supervised Principal Component Analysis (SPCA) combined with multinomial logistic regression and LASSO multinomial logistic regression for multiclass cancer classification using high-dimensional RNA-Seq gene expression data from the TCGA PANCAN dataset ($n$ = 801, $p$ = 20,531). Both methods were evaluated on a stratified hold-out test set of 238 observations covering five cancer types: BRCA, COAD, KIRC, LUAD, and PRAD.

Both models achieved identical and perfect classification performance, with 100% overall accuracy, a macro F1-score of 1.0000, and a multiclass AUC of 1.0000 on the test set. All per-class metrics, including sensitivity, specificity, precision, and F1-score, were also equal to 1.0000 for every cancer type under both methods. Based on these results, neither method showed better predictive performance on this dataset.

The two methods differed substantially in their feature reduction strategies. LASSO directly selected 33 genes from the full 20,531 gene feature space and used these genes as predictors. This provides a compact and interpretable gene-level feature set. In contrast, SPCA first selected 100 genes through ANOVA based screening and then compressed them into 6 principal components, producing the smaller final model input. Therefore, LASSO selected fewer original genes, while SPCA produced a more compact final representation.

The choice between these two methods should depend on the main goal of the analysis. LASSO is more suitable when gene-level interpretability and biomarker identification are important, because it directly identifies individual genes with non-zero coefficients. SPCA is more suitable when the goal is to obtain a low-dimensional representation for prediction, although the resulting principal components are less directly interpretable.

Future work should evaluate these methods on more challenging multiclass problems, such as cancer subtype classification within the same tissue type, where the molecular differences between classes are less clear. Biological validation of the genes selected by LASSO and further analysis of the SPCA component loadings would also strengthen the practical relevance of these findings.

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